

Editing and Reading Data from Files – Reading and Writing Files in R

Programming

Name: Savitha R

Reg.No: 192421368

R provides several functions to read, write, and edit data from various file formats. This is

essential for data analysis, preprocessing, and reporting.

1. Reading Data from Files

R can read multiple file types:

1. Reading CSV Files:

data <- read.csv("file.csv", header = TRUE)

output:

```
  Name Marks
1 Asha    85
2 Ravi    90
```

2. Reading Text Files:

data <- read.table("file.txt", header = TRUE, sep =
"\\t")

output:

```
> print(data)
  Name Marks
1 Asha    85
2 Ravi    90
```

3. Reading Excel Files:

library(readxl)

data <- read_excel(""file.xlsx")

output:

```
> data <- read_excel("E:/New folder/student.xlsx")
> print(data)
# A tibble: 2 × 2
  Name Marks
  <chr> <dbl>
1 Asha    85
2 Ravi    90
>
```

4. Reading RDS Files:

data <- readRDS(""file.rds")

output:

```
> student <- data.frame(
+   Name = c("Asha", "Ravi"),
+   Marks = c(85, 90)
+ )
>
> saveRDS(student, "student.rds")
> data <- readRDS("student.rds")
> print(data)
  Name Marks
1 Asha    85
```

5. Reading JSON Files:

library(jsonlite)

data <- fromJSON(""file.json")

output:

```
> library(jsonlite)
> library(jsonlite)
>
> data <- fromJSON("E:/New folder/student.json")
>
> print(data)
  Name Marks
1 Asha    85
2 Ravi    90
```

2. Writing Data to Files

1. Writing CSV Files:

`write.csv(data, "output.csv", row.names = FALSE)`

output:

```
> data <- data.frame(
+   Name = c("Asha", "Ravi"),
+   Marks = c(85, 90)
+ )
>
> print(data)
  Name Marks
1 Asha    85
2 Ravi    90
>
> write.csv(data, "E:/New folder/output.csv", row.names = FALSE)
> newdata <- read.csv("E:/New folder/output.csv")
> print(newdata)
  Name Marks
1 Asha    85
2 Ravi    90
```

2. Writing Text Files:

`write.table(data, "output.txt", sep = "\t", row.names = FALSE)`

output:

```
> data <- data.frame(
+   Name = c("Asha", "Ravi"),
+   Marks = c(85, 90)
+ )
>
> print(data)
  Name Marks
1 Asha    85
2 Ravi    90
> write.table(data,
+             "E:/New folder/output.txt",
+             sep = "\t",
+             row.names = FALSE)
> newdata <- read.table("E:/New folder/output.txt",
+                       header = TRUE,
+                       sep = "\t")
>
> print(newdata)
  Name Marks
1 Asha    85
2 Ravi    90
```

3. Writing Excel Files:

`library(writexl)`

`write_xlsx(data, "output.xlsx")`

output:

```

> library(writexl)
> data <- data.frame(
+   Name = c("Asha", "Ravi"),
+   Marks = c(85, 90)
+ )
>
> print(data)
  Name Marks
1 Asha    85
2 Ravi    90
> write_xlsx(data, "E:/New folder/output.xlsx")

```

4. Writing RDS Files:

`saveRDS(data, "output.rds")`

output:

```

> data <- data.frame(
+   Name = c("Asha", "Ravi"),
+   Marks = c(85, 90)
+ )
>
> print(data)
  Name Marks
1 Asha    85
2 Ravi    90
> saveRDS(data, "E:/New folder/output.rds")
> newdata <- readRDS("E:/New folder/output.rds")
>
> print(newdata)
  Name Marks
1 Asha    85
2 Ravi    90

```

5. Writing JSON Files:

`library(jsonlite)`

`write_json(data, "output.json")`

output:

```

> library(jsonlite)
> data <- data.frame(
+   Name = c("Asha", "Ravi"),
+   Marks = c(85, 90)
+ )
>
> print(data)
  Name Marks
1 Asha    85
2 Ravi    90
> write_json(data, "E:/New folder/output.json")
> library(jsonlite)
>
> newdata <- fromJSON("E:/New folder/output.json")
>
> print(newdata)
  Name Marks
1 Asha    85
2 Ravi    90

```

3. Editing Data in R

Editing data involves modification using R functions:

- Adding Columns:

`data$new_col <- c(1,2,3)`

output:

```
> data <- data.frame(
+   Name = c("Asha", "Ravi", "Kiran"),
+   Marks = c(85, 90, 88)
+ )
> 
> print(data)
  Name Marks
1  Asha   85
2  Ravi   90
3 Kiran   88
> data$new_col <- c(1, 2, 3)
> 
> print(data)
  Name Marks new_col
1  Asha   85      1
2  Ravi   90      2
3 Kiran   88      3
> data <- data.frame(
+   Name = c("Asha", "Ravi", "Kiran"),
+   Marks = c(85, 90, 88)
+ )
> 
> data$new_col <- c(1, 2, 3)
> 
> print(data)
  Name Marks new_col
1  Asha   85      1
2  Ravi   90      2
3 Kiran   88      3
```

- Removing Columns:

`data$col <- NULL`

output:

```

> data <- data.frame(
+   Name = c("Asha", "Ravi", "Kiran"),
+   Marks = c(85, 90, 88),
+   Grade = c("A", "A+", "B+")
+ )
>
> print(data)
  Name Marks Grade
1 Asha    85     A
2 Ravi    90    A+
3 Kiran   88    B+
> data$Grade <- NULL
>
> print(data)
  Name Marks
1 Asha    85
2 Ravi    90
3 Kiran   88

```

- Renaming Columns:

names(data)[names(data) == "old"] <- "new";

output:

```

> data <- data.frame(
+   Name = c("Asha", "Ravi", "Kiran"),
+   Marks = c(85, 90, 88)
+ )
>
> print(data)
  Name Marks
1 Asha    85
2 Ravi    90
3 Kiran   88
> names(data)[names(data) == "Marks"] <- "score"
>
> print(data)
  Name score
1 Asha    85
2 Ravi    90
3 Kiran   88

```

- Filtering Rows:

data <- subset(data, age > 20)

output:

```

> data <- data.frame(
+   Name = c("Asha", "Ravi", "Kiran"),
+   Age = c(18, 22, 25)
+ )
>
> print(data)
  Name Age
1 Asha  18
2 Ravi  22
3 Kiran  25
> data <- subset(data, Age > 20)
>
> print(data)
  Name Age
2 Ravi  22
3 Kiran  25

```

- Updating Values:

`data$Age[data$Age < 18] <- 18`

output:

```

> data <- data.frame(
+   Name = c("Asha", "Ravi", "Kiran"),
+   Age = c(16, 20, 15)
+ )
>
> print(data)
  Name Age
1 Asha  16
2 Ravi  20
3 Kiran  15
> data$Age[data$Age < 18] <- 18
>
> print(data)
  Name Age
1 Asha  18
2 Ravi  20
3 Kiran  18
> data <- data.frame(
+   Name = c("Asha", "Ravi", "Kiran"),
+   Age = c(16, 20, 15)
+ )
>
> data$Age[data$Age < 18] <- 18
>
> print(data)
  Name Age
1 Asha  18
2 Ravi  20
3 Kiran  18

```

4. Useful Tips and Best Practices in R (RStudio)

These are useful commands and examples.

1. Check the Current Working Directory

Program

```
getwd()
```

Output (Example)

```
[1] "E:/New folder"
```

2. Change the Working Directory

Program

```
setwd("E:/New folder")
```

Verify

```
getwd()
```

Output

```
[1] "E:/New folder"
```

3. Prevent Automatic Factor Conversion

Program

```
data <- data.frame(  
  Name = c("Asha", "Ravi"),  
  Grade = c("A", "B"),  
  stringsAsFactors = FALSE  
)
```

```
print(data)
```

Output

```
  Name Grade  
1 Asha    A  
2 Ravi    B
```

4. Read Large Files Faster (data.table)

Install the package (only once)

```
install.packages("data.table")
```

Program

```
library(data.table)
```

```
data <- fread("E:/New folder/student.csv")
```

```
print(data)
```

Output (Example)

Name Marks

1: Asha 85

2: Ravi 90

5. Write a CSV File Using write.csv2()

Program

```
data <- data.frame(  
  Name = c("Asha", "Ravi"),  
  Marks = c(85, 90)  
)
```

```
write.csv2(data, "E:/New folder/output.csv", row.names = FALSE)
```

Result: The file **output.csv** is created in **E:\New folder**.

Summary

Function	Purpose
getwd()	Displays the current working directory
setwd()	Changes the working directory

Function	Purpose
stringsAsFactors = FALSE	Keeps text columns as character values
fread()	Reads large CSV files quickly
write.csv2()	Writes CSV files using regional formatting (e.g., semicolon separators where commas are decimal separators)